fn make_bounded_range_to_pureDP

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn make_bounded_range_to_pureDP`. This proof is an adaptation of Lemma 2 here, which proves the conversion between bounded range [**?**] and pure DP.

1 Hoare Triple

Preconditions

Compiler-verified

- Variable `eta` must be of type float64

User-verified

None

Pseudocode

```
def make_bounded_range_to_pureDP(meas: Measurement) -> Measurement:
    return meas.with_map(
        meas.input_metric,
        ZeroConcentratedDivergence.default(),
        PrivacyMap.new_fallible(lambda d_in: meas.privacy_map(d_in)),
    )
```

Postcondition

Definition 1.1 (η-Range Bounded)**.** Let $M : X^n \to Y$ be a randomized mechanism and let $\eta > 0$. We say that M is ε-range-bounded if $\forall y \in Y$ and any pair of neighboring datasets $x, x' \in X^n$, there exists a t such that:

$$\log\left(\frac{\Pr[M(x) = y]}{\Pr[M(x') = y]}\right) \in [t, t + \eta]\,. \tag{1}$$

Definition 1.2 (ε-Differential Privacy)**.** Let $M : X^n \to Y$ be a randomized mechanism and let $\varepsilon > 0$. We say that M is ε-differentially private if for any pair of neighboring datasets $x, x' \in X^n$, and $\forall y \in Y$, it holds that

$$\left|\log\left(\frac{\Pr[M(x) = y]}{\Pr[M(x') = y]}\right)\right| \leq \varepsilon \cdot . \tag{2}$$

Bounded range demans that the privacy loss distribution (PLD) has a support of length η, while differential privacy demands that the privacy loss is bounded by *epsilon*. PLDs cross 0 (they cannot be only positive or negative) and $t > -\eta$, meaning that the support crosses 0. This implies that $[t, t + \eta] \in [-\eta, \eta]$. This implies that η-Bounded Range mechanisms satisfy ϵ-Differential Privacy by definition.